115學年度大學部專題海報展



國立清華大學 資訊工程學系

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Beautified Is Not Forged

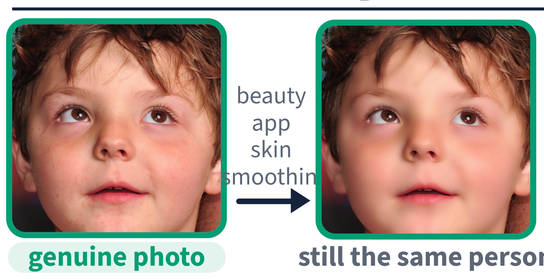
A face checker that answers **real**, **beautified** or **forged**, and marks the evidence so you can check it.

Qin-Ying Lin · Bo-Rong Chen · Chen-Shan Yu NTHU Capstone Project 2026

16/17 published deepfake detectors call a smoothed genuine face **forged**. Ours does not.

1 One beauty edit, two detector errors

scores at each detector's own threshold (5% false alarms on untouched faces)



Prior detectors · real / fake only

FAKE false alarm

Effort	Forensics Ad.	GenD	DFD-HR
0.72	0.95	0.97	0.94

Ours · real / beautified / forged

BEAUTIFIED

real | **beautified** | forged



Prior detectors · real / fake only

GENUINE missed

Effort	Forensics Ad.	GenD	DFD-HR
0.44	0.67	0.70	0.74

Ours · the forgery score never reads the beauty score

FORGED

real | beautified | **forged**

13 / 17 detectors call **36–69%** of smoothed genuine faces forged: a real / fake detector has no third answer.

2 What a beauty filter does

one FFHQ face, four operations of a retouching service our model never saw in training (Alibaba, strength 90)

Commercial apps apply several of these at once and never say where. We split their renders into single operations with an exact footprint, and train on the pieces. Below each face: how often our model calls that operation **beautified**.



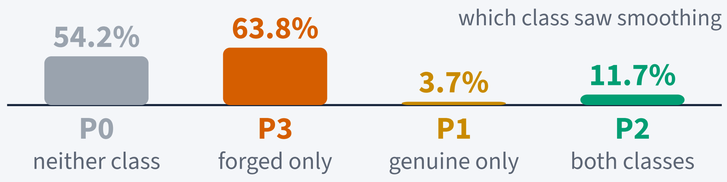
original	eye enlarging	face slimming	skin smoothing	whitening
78.4% called real	**69.7%** beautified	**61.3%** beautified	**98.8%** beautified	**72.3%** beautified

same model, same data, same recipe: only which class saw smoothing in training changes

3 Why it happens, and what we built

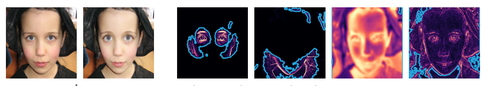
The error is set in training

smoothed genuine faces called forged (%) at 5% false alarms · report Table: exposure



One backbone, three supervision sources, four heads

Paired edits with an exact footprint M



one render → eye · contour · tone · texture



18,734 renders → 50,112 single operations
61,510 part-level forgeries
41,512 FF++ frames, both classes smoothed

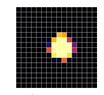
CLIP ViT-L/14 + LoRA



24 blocks frozen (303 M)
LoRA r = 16 on q, k, v, o
3.16 M parameters trained

Four heads, one checked explanation

classification	real / forged / beautified
operation	eye, contour, tone, texture
quantity	auxiliary amounts
evidence	16×16 map, trained on M



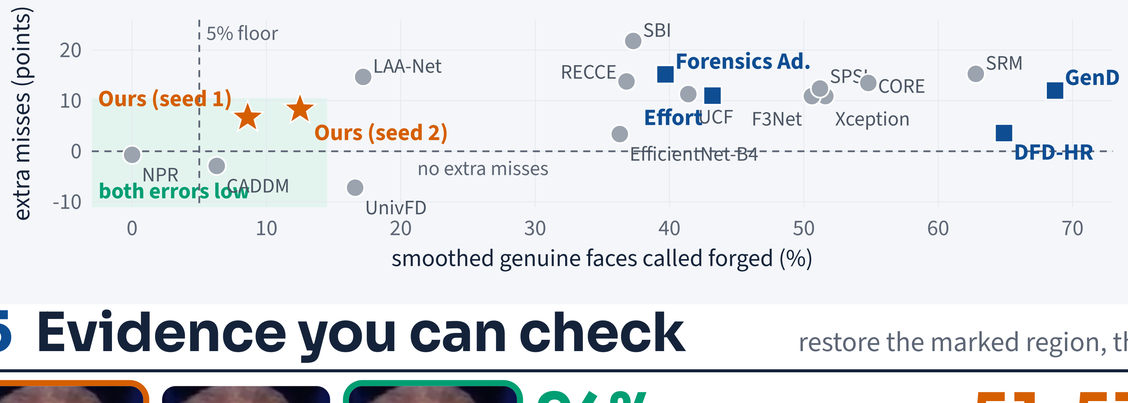
evidence map

"The evidence is concentrated on the nose (0.99)."
code composes it, rules check it, no language model

4 Against 17 published detectors

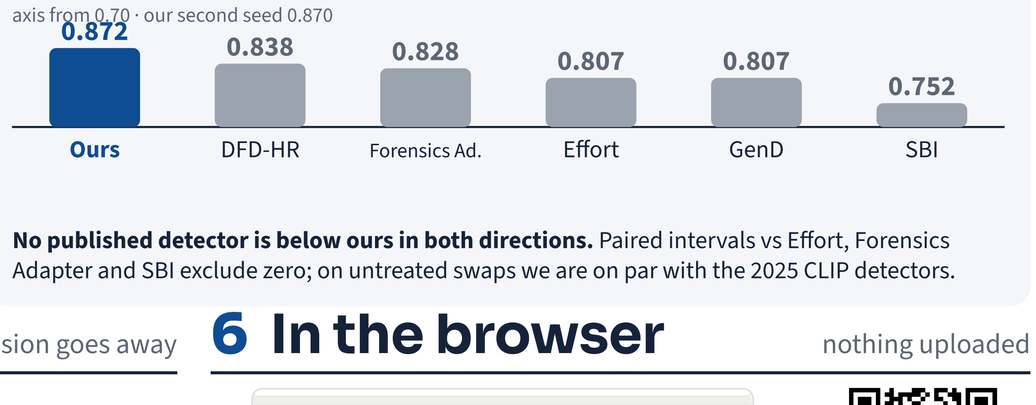
official weights, same frames, same crops · FF++ test recall real 82.7 · forged 90.9 · beautified 86.7%

Both errors at once · lower left is better



Same training videos, after beautification

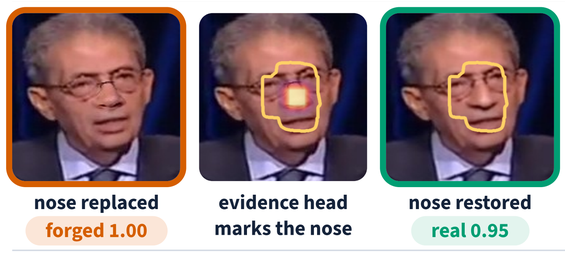
Celb-DF-B video AUC; every model trained on the same FF++ videos



No published detector is below ours in both directions. Paired intervals vs Effort, Forensics Adapter and SBI exclude zero; on untreated swaps we are on par with the 2025 CLIP detectors.

5 Evidence you can check

restore the marked region, the decision goes away



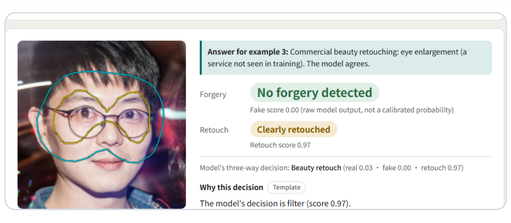
nose replaced forged 1.00 | **evidence head marks the nose** | **nose restored** real 0.95

94% of forgery decisions removed by restoring the marked region (ceiling 94.1%); Grad-CAM 51%

51–53% VLM descriptions matched to their own photo blind: chance level, so code writes every sentence (361/361 pass)

6 In the browser

nothing uploaded



try it on your phone

0.59 s per photo

Limits. Reenactment is not covered (AUC 0.50 / 0.56) · 20.8% of untouched originals are called beautified · monochrome filters are the weakest · across services only smoothing is named reliably.

github.com/Doreen1113/beautified-not-forged · code, scores of 17 detectors, benchmark manifest, demo